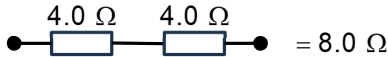
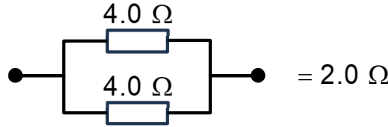
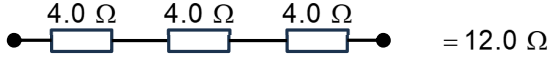
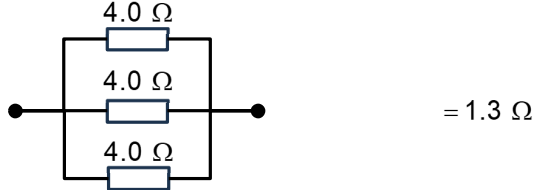
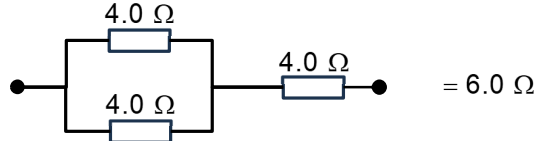


		R_{max} $\rightarrow$ lamp $\rightarrow$ variable resistor
21	B	In order for the voltmeter to have a reading of 12 V, the resultant resistance of component C, $R_C = 6.0 \, \Omega$, so by potential divider principle, $V_C = \frac{6.0}{6.0 + 6.0} \times 24 = 12 \, \text{V}.$ Option A: Incorrect   Option B: Correct    Options C and D will no longer be valid since the question asks for the least number of $4.0 \, \Omega$ resistors.

Qn	Answer	Solution
22	A	When temperature of thermistor is 50°C $R = 1.00\text{ k}\Omega$